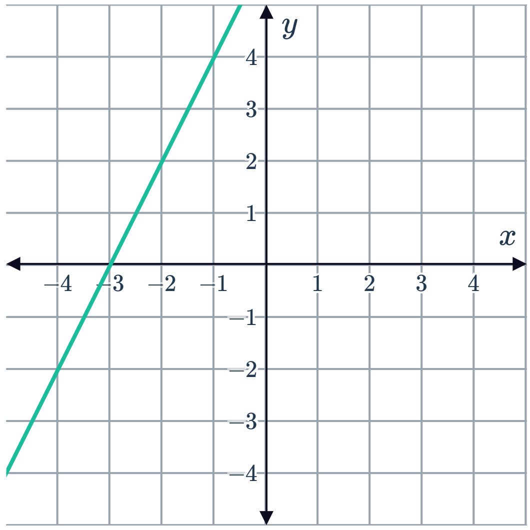
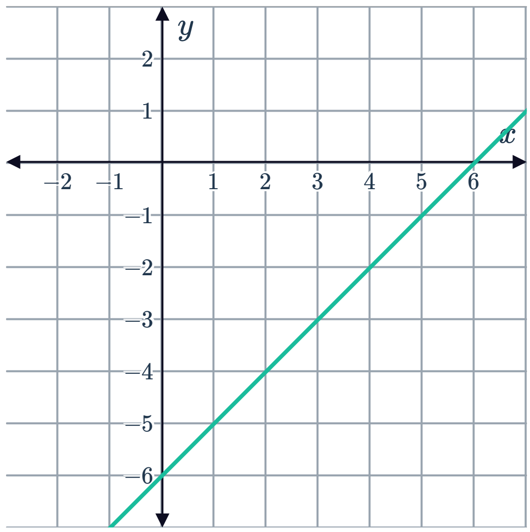
Linear equations and inequalities

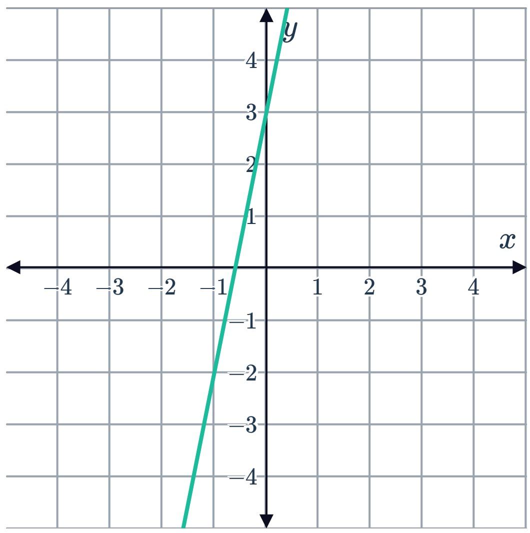
1 Consider the graph of $y = 2x + 6$. Using the graph, solve the inequality $2x + 6 \geq 0$.



2 Consider the graph of $y = x - 6$. Using the graph, solve the inequality $x - 6 < 0$.



3 Consider the graph of $y = 5x + 3$. Using the graph, solve the inequality $5x + 3 \leq 0$.



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4 Consider the equation $2(x-1)-3=7$.

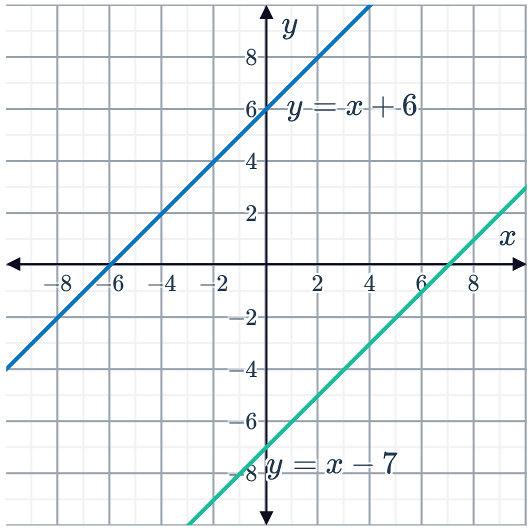
a Solve for the value of x that satisfies the equation.

b To verify the solution graphically, what two straight lines need to be graphed?

c Graph these lines on the same number plane.

d Hence find the value of x that satisfies the two equations.

5 Consider the graphs of $y=x+6$ and $y=x-7$:
How many solutions does the inequality $x+6 \geq x-7$ have?



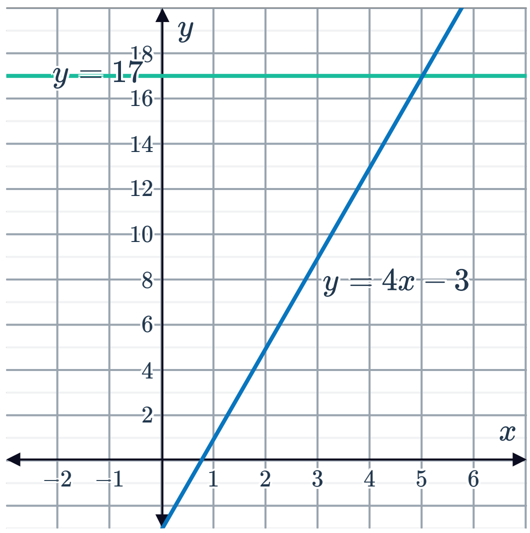
6 Consider the inequality $2x-4>2-4x$.

a Sketch the graphs of the lines for $y=2x-4$ and $y=2-4x$.

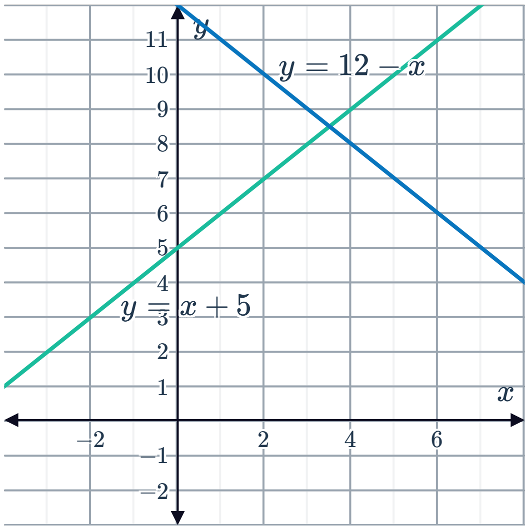
b Find the point of intersection of the lines.

c Hence, solve the inequality $2x-4>2-4x$.

7 Consider the graph of the lines $y=17$ and $y=4x-3$:
Using the graphs, solve the inequality $4x-3<17$.



8 Consider the graphs of $y = x + 5$ and $y = 12 - x$:
Using the graphs, solve the inequality $x + 5 > 12 - x$.



9 Use sign diagrams to solve the following inequalities:

a $5x - 15 > 0$ b $8x + 32 \leq 0$ c $-3x + 81 \geq 0$ d $-20x > 80$

Quadratic equations and inequalities

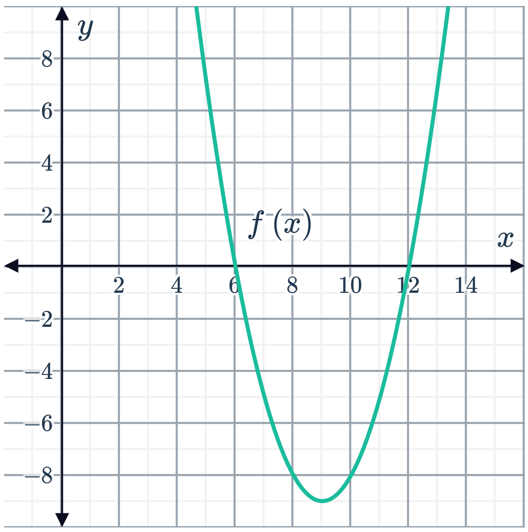
10 Consider the graph of $y = f(x)$:

a Find the values of x for which $f(x) = 0$.

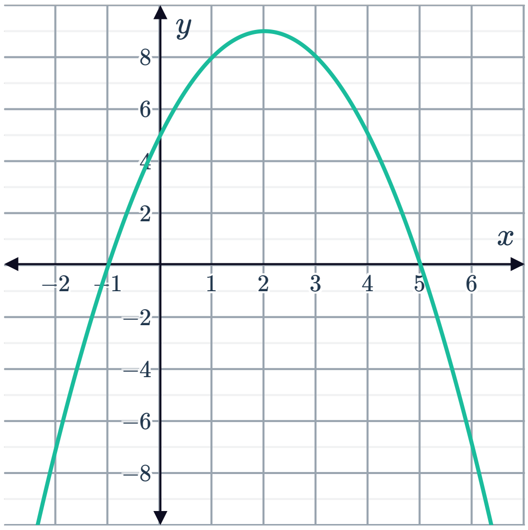
b For what values of x is $f(x) < 0$?

c For what values of x is $f(x) > 0$?

d What is the x-coordinate of the vertex of $f(x)$?



11 Consider the function $f(x) = 5 + 4x - x^2$.
Use the graph to solve the inequality
$5 + 4x - x^2 > 0$.



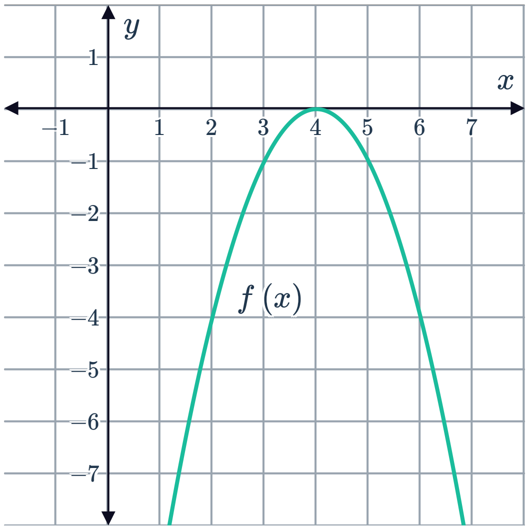
12 Consider the graph of $y = f(x)$:

a For what values of x is $f(x) < 0$?

b For what value of x is $f(x) \geq 0$?

c What is the axis of symmetry of $f(x)$?

d What is the value of the discriminant of $f(x)$?



13 Consider the function $f(x) = x^2 - 4x - 5$.

a Sketch the graph of the function.

b Hence state the values of x for which $f(x) \leq 0$.

14 Consider the function $f(x) = 3x^2 - 2x - 8$.

a Solve the equation $f(x) = 0$.

b Sketch the graph of the function.

c Hence state the values of x for which $f(x) \geq 0$.

15 Consider the inequality $(x-3)^2 \leq 0$.

a How many x-intercepts does the graph of $y = (x-3)^2$ have?

b Solve the inequality.

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16 Consider the function $y = 2x^2 + 9x + 8$.

a Determine the x-intercepts of the function.

b Is the graph concave up or concave down?

c Hence find the values of x for which $y > 0$.

17 Consider the function $f(x) = x^2 - 2x$.

a Sketch the graph of the function.

b Hence state the values of x for which $f(x) \leq 8$.

18 Consider the inequality $x^2 - 2x \leq -x + 2$.

a Sketch the graphs of $y = x^2 - 2x$ and $y = -x + 2$ on the same number plane.

b State the x-values for the points of intersection.

c Hence solve the inequality $x^2 - 2x \leq -x + 2$.

19 Consider the inequality $x^2 > 6x - 5$.

a Sketch the graphs of $y = x^2$ and $y = 6x - 5$ on the same number plane.

b State the x-values for the points of intersection.

c Hence solve the inequality $x^2 > 6x - 5$.

20 Consider the inequality $3x^2 + x \geq 2x^2 + 2$.

a Sketch the graphs of $y = 3x^2 + x$ and $y = 2x^2 + 2$ on the same number plane.

b Hence or otherwise, solve the inequality $3x^2 + x \geq 2x^2 + 2$.

21 Use sign diagrams to solve the following inequalities:

a $x^2 - 9 > 0$

b $2x^2 - 4x \leq 0$

c $x^2 - 5x + 6 \geq 0$

d $x^2 + 7x + 12 > 0$

Equivalent inequalities

22 To solve the inequality $x \leq 2x - 3$, Christa graphed $y = x + 3$. What other line would she need to graph to be able to solve the inequality graphically?

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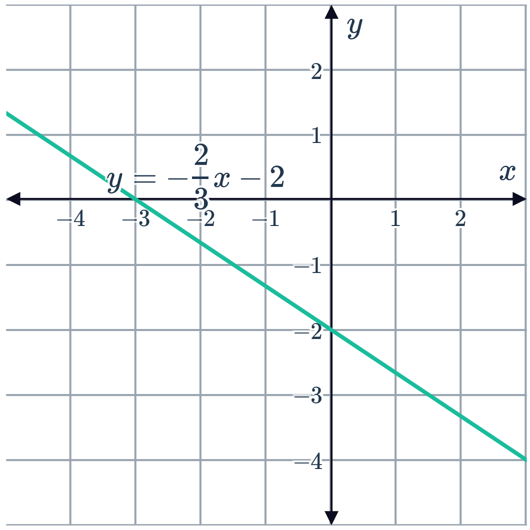
23 To solve the inequality $x \leq \frac{x-3}{4} - 1$, Tra[redacted]phed $y = x - 3$. What other line would she need to graph to be able to solve the inequality graphically?

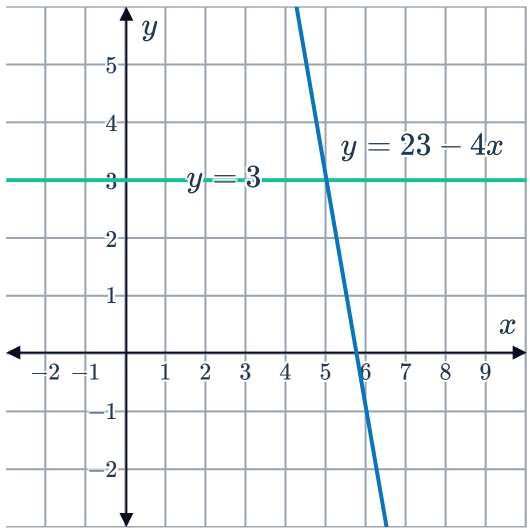
24 Consider the graph of $y = -\frac{2}{3}x - 2$.
Using the graphs, solve the inequality $-2x - 6 > 0$.



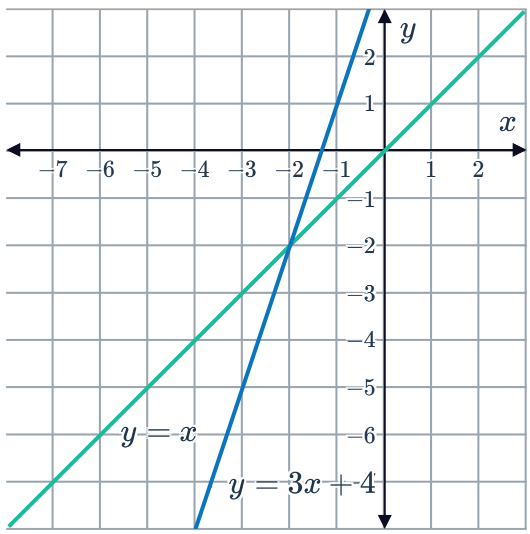
25 Consider the graph of the lines $y = 3$ and $y = 23 - 4x$:

a Using the graphs, solve the inequality $23 - 4x < 3$.

b Using the graphs, solve the inequality $-20 + 4x \geq 0$.



26 Consider the graphs of $y = 3x + 4$ and $y = x$:
Using the graphs, solve the inequality $3x + 4 - x \leq 0$.



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27 Explain whether the following inequalities [redacted]e solved graphically using the graphs of:

$$f(x) = 4x + 3 \text{ and } g(x) = 5 - x$$

a $(4x + 3) - (5 - x) \leq 0$

b $(4x + 3) + (3 - x) < 0$

c $\dfrac{4x + 3}{5} + x > 0$

d $3x + 8 < 0$

e $5x > 2$

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